

Phosphorus



Sulpher

Non - metals

Table 3.3 Difference between metals and non-metals

Metals	Non-Metals
Metals are lustrous. They have a shiny surface.	Non metals are non lustrous. They have non- shiny surface.
Metals are generally hard.	Non-metals are generally soft.
Most metals can be bent, beaten into sheets and they can be drawn into wires.	Non-metals can not be bent, beaten into sheets and they can not be drawn into wires.
Most metals are good conductors of electricity.	Non-metals are bad conductors of electricity.
Most metals are good conductors of heat.	Non-metals are bad conductors of heat.
Most metals make ringing sound when struck. Hence, they are used to make objects like bells.	Non-metals does not make any sound when they are struck.

Metalloids

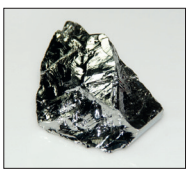
Metalloids exhibit the properties of both metals and non metals. Silicon, arsenic, antimony, and boron are some examples of metalloids.



Boron (B)



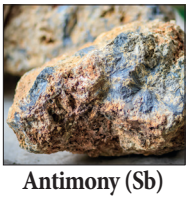
Silicon (Si)



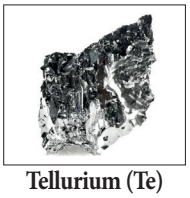
Germanium (Ge)



Arsenic (As)



Antimony (Sb)



Tellurium (Te)

Metalloids

3.3.2 Symbol of an element

A symbol is an abbreviation or short representation of a chemical element. There is a unique symbol for each element. It represents one atom of the element. The symbol is usually derived from the name of the element, which is either in English or Latin. These symbols are accepted by the International Union of Pure and Applied Chemistry (IUPAC).

Dalton was the first scientist to use the symbols for elements in a very specific sense. When he used a symbol for an element he also meant a definite quantity of that element, that is, one atom of that element. Berzelius suggested that the symbols of elements can be written as one or two letters of the name of the element.

The following rules are followed while assigning symbol to an element.



In the beginning, the names of elements were derived from the name of the place where they were found for the first time. For example, the name copper was taken from Cyprus. Some names were taken from specific colours. For example, gold was taken from the English word meaning yellow. Now-a-days, IUPAC approves names of elements. Many of the symbols are the first one or two letters of the element's name in English. The first letter of a symbol is always written as a capital letter (uppercase) and the second letter as a small letter (lowercase).